Arushi

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Skills

- **Programming and Databases**: Python, Java, MySQL, PostgreSQL
- **Data Analysis & Visualization**: NumPy, Pandas, Matplotlib, Seaborn, Plotly
- **Data Science and ML** scikit-learn, TensorFlow, keras, FastAPI
- **Data Engineering**: Apache Airflow, Astronomer, Docker
- **Soft Skills**: Critical Thinking, Self-Driven Learning, Clear Communication

Projects

Real Time Space Weather Analysis and Prediction September 2025 – December 2025

- Collected and analyzed 16 years of historical space weather data (320MB, 500K+ events), performing time-series analysis to identify patterns in solar activity, discovering CMEs preceded geomagnetic storms by 2-3 days with 70% correlation
- Implemented Isolation Forest and PCA for anomaly detection, identifying unusual solar activity with 75% precision, improving accuracy by 25% over threshold methods
- Generated an automated Airflow pipeline and Streamlit dashboard for real-time monitoring

Tech Stack: MySQL, Python, scikit-learn, Airflow, Docker, Git

Real Time Stock Analytics Engine: June 2025 – July 2025

- Investigated real-time price movements for 50+ stocks with 10-second updates, computing rolling volatility and cross-market correlation patterns
- Engineered optimized data structures achieving O(1) query performance for min/max tracking and automated alert generation
- Created a Streamlit dashboard visualizing intraday trends with automated alerts for significant price movements

Tech Stack: Python, Pandas, Apache Airflow, Dash, MySQL

Trending YouTube Videos Analysis & Virality Prediction: February 2025 – April 2025

- Analyzed 10,000+ trending videos across 5 countries, revealing videos posted 12-3pm had 40% higher engagement and identifying key factors driving virality across different regions
- Built Random Forest classifier predicting video virality with 78% accuracy, validated through k-fold cross-validation and maintained 74% accuracy on unseen country data.
- Designed interactive Tableau dashboard for exploring engagement trends and providing data-driven content strategy recommendations

Tech Stack: Python, Pandas, NumPy, scikit-learn, YouTube API, Tableau

Certificates

- NPTEL Python for Data Science [Silver + Elite] 🔗 August 2025
- Mastering Data Structures and Algorithms (Lovely Professional University 🔗 August 2025
- Python Programming Course (iHub DivyaSampark, IIT Roorkee) September 2024

Achievements

- **Top 15 in Internal Hackathon for Smart India Hackathon**:
  Gear Up Season 4: 36-hour hackathon winners among more than 450 teams
- **One among Dean's top 10 % students at university**:
  For good academic performance and extracurricular activities at university.

Education

- **Lovely Professional University** Phagwara, Punjab, India
  Bachelor of Technology - Computer Science and Engineering; ***CGPA: 8.81*** *Since August 2023*
- **D.A.V. Public School** Punjab, India
  Class X; Percentile: 87.8% *2021*
  Class XII; Percentile: 88.2% *2023*